AP PRECALCULUS

Unit 1 • Worksheet 1.2B

Worksheet 1.2B • Rates of Change

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The table gives values of g. What is the average rate of change of g on $[2, 8]$?

x	2	4	6	8
$g(x)$	-3	1	6	9

(A) 12

(B) 2

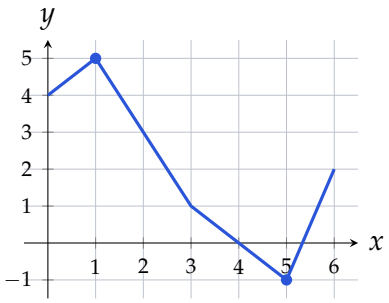
(C) $\frac{1}{2}$

(D) 6

Question 2

NO CALCULATOR

The graph of h consists of line segments. What is the average rate of change of h on $[1, 5]$?



(A) $-\frac{3}{2}$

(B) -6

(C) $\frac{3}{2}$

(D) -1

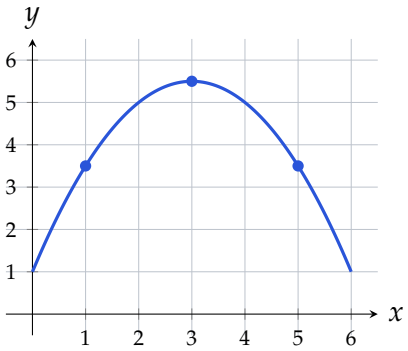
Question 3 NO CALCULATOR

Using the same graph of h, on which interval is the average rate of change of h equal to -1?

(A) $[0, 3]$
(B) $[3, 6]$
(C) $[1, 3]$
(D) $[0, 6]$

Question 4 (Free response) NO CALCULATOR

The graph of f is shown.



(a) Find the average rate of change of f on $[1, 3]$ and on $[3, 5]$.
(b) Draw the secant line for $[1, 5]$. What is its slope?
(c) Is the rate of change of f increasing or decreasing on $[0, 6]$? What does this say about concavity?